

Following completion of the preparation materials, students should be able to:

**1. PK properties:**

Compare the class pharmacokinetic properties of the direct-acting muscarinic and nicotinic cholinomimetics to those of the individual agents, including the tendency to cross the blood-brain barrier and the effect of pH on excretion in urine.

**2. Physiology-Pharmacology relationship → Mechanism of action and effects:**

Relate the ANS neurotransmitter chemistry, receptor characteristics, and functional integration to the structure-activity relationship, mechanism of action, and effects of each of the direct-acting cholinomimetics.

**3. PK properties + Mechanism of action + Effect → Mechanisms of Drug interactions**

Apply the pharmacokinetic properties, therapeutic effects, adverse effects, and potential drug-disease and drug-drug interactions of the individual agents to the selection of drug therapy for the individual patient.

**4. PK + Effects + Drug-disease and Drug-drug interactions → Cautious use:**

Predict the cautions and contraindications for use based on mechanisms of action, adverse effects, and drug interactions.

**5. Poisoning:**

Correlate the signs and symptoms of nicotinic toxicity with the mechanisms of action and the effects of the drugs.

## Key points: Definitions

|                                     |                                                                                                                                                                                                                      |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetylcholine (ACh)                 | Neurotransmitter of the ANS acting at the ganglia and effector organs of the parasympathetic nervous system (PNS) and sweat glands                                                                                   |
| Acetylcholinesterase (AChE)         | Enzyme that degrades acetylcholine in the synapse                                                                                                                                                                    |
| Norepinephrine (NE)                 | Neurotransmitter of the ANS at the effector organs of the sympathetic nervous system (SNS), also secreted by the adrenal medulla                                                                                     |
| Epinephrine (Epi)                   | Hormone of the ANS secreted by the adrenal medulla and acting at the effector organs of the SNS                                                                                                                      |
| Ionotropic receptors                | Membrane-bound receptor proteins that respond to ligand binding by opening an ion channel and allowing ions to flow into the cell, either increasing or decreasing the likelihood that an action potential will fire |
| G protein-coupled receptors (GPCRs) | Membrane-bound receptor proteins that, when activated by a ligand, activate enzymes (second messengers), operate ion channels, and initiate intracellular cascades                                                   |
| Metabotropic receptors              | GPCRs that indirectly link to ion channels through a signaling mechanism involving second messengers                                                                                                                 |

## Key points: Acetylcholine

- All preganglionic efferent autonomic nerve fibers and somatic motor fibers to skeletal muscle, all parasympathetic postganglionic fibers and sympathetic fibers innervating sweat gland release the neurotransmitter **acetylcholine** – these are **cholinergic neurons**.
- Acetylcholine is synthesized in the cytosol from acetyl-CoA and choline and stored in vesicles. It is released into the synapse when an action potential reaches the nerve terminal and triggers sufficient influx of calcium ions via N-type calcium channels.
- Acetylcholine is degraded in the synaptic cleft by acetylcholinesterase.
- Acetylcholine activates nicotinic and muscarinic receptors in the CNS and periphery:
  - preganglionic nicotinic receptors (ionotropic receptors)
  - neuromuscular nicotinic receptors (ionotropic receptors)
  - muscarinic G protein-coupled receptors (metabotropic receptors) on effector organs

## Key points: Acetylcholine receptors and actions of cholinomimetic drugs

- Nicotinic receptors (nAChRs) are ligand-gated nonselective cation channels permeable to  $\text{Na}^+$  and  $\text{K}^+$ ; some are permeable to  $\text{Ca}^{2+}$ . Two molecules of ACh binding at the interface between the  $\alpha$  subunits and the adjacent subunits open the nicotinic receptor ion channel, permitting one or more specific ions to diffuse through.
- Five subgroups of muscarinic receptors, mChRs, ( $M_{1-5}$ ) have been described. They are:
  - $M_1$ ,  $M_3$ ,  $M_5$  couple to the  $G_{q/11}$ PCR family of G proteins.
  - $M_2$ ,  $M_4$  preferentially couple to the  $G_{i/o}$ PCR family of G proteins.
- Cholinomimetic drugs – acetylcholine-like drugs – have numerous and complex actions.
- A drug's structure determines its relative selectivity for receptor type – muscarinic or nicotinic receptors – and resistance to hydrolysis – **the structure-activity relationship**.

## Key points: Properties of direct-acting cholinomimetics

- The direct-acting cholinomimetics are choline esters and naturally occurring alkaloids that bind specific receptor types and trigger an action potential or activate a second messenger signaling system of that receptor subtype. That is, they are nicotinic or muscarinic receptor agonists.
- The choline esters, which are quaternary ammonium compounds, are poorly absorbed from the GI tract and have limited ability to cross the blood-brain barrier. They are short-acting and rapidly eliminated by the kidneys.
- The alkaloids, such as nicotine and pilocarpine, are weak bases and tertiary amines. They are readily absorbed and cross the blood-brain barrier. They are metabolized by the liver. The metabolites are primarily eliminated by the kidney.
- The various agents are relatively nonselective and differ in their spectrums of action – that is, the degree of muscarinic versus nicotinic stimulation and in their relative affinities for the muscarinic receptor subtypes.

## Key points: Direct-acting cholinomimetics' clinical uses and adverse effects

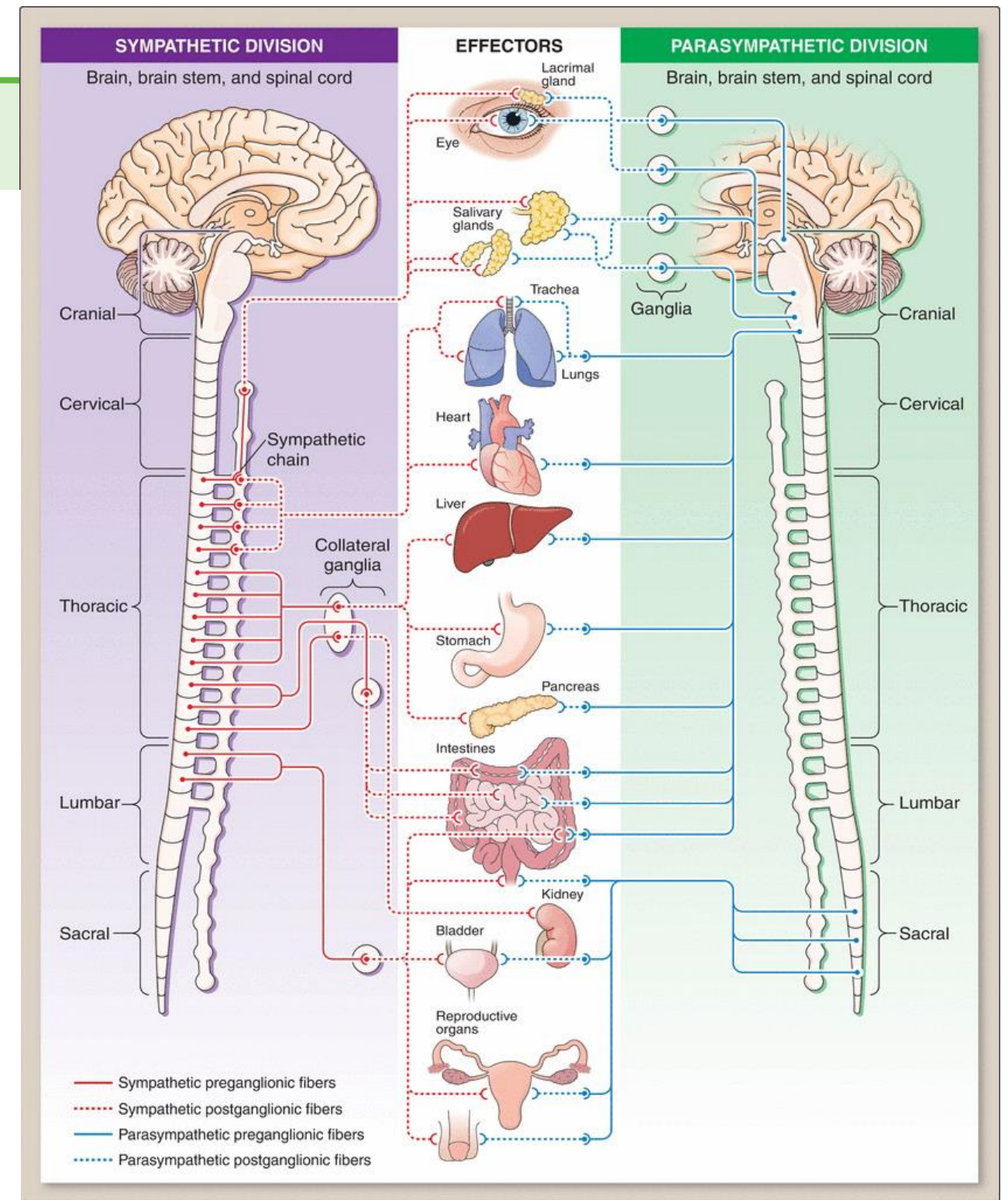
- The various direct-acting cholinomimetics are used in the treatment of urinary bladder disorders, xerostomia, glaucoma, miotic agents, and in the diagnosis of bronchial hyperreactivity.
- Muscarinic toxicity is characterized by CNS stimulation (uncommon with choline esters, which do not cross BBB), miosis, ciliary muscle spasm (spasm of accommodation), bronchoconstriction, excessive gastrointestinal and genitourinary smooth muscle activity, increased secretory activity (sweat glands, airway, gastrointestinal tract, lacrimal glands), and vasodilation.
- Nicotinic receptor agonists are smoking cessation aids.
- Nicotine's toxic effects include ganglionic stimulation and depolarizing blockade and neuromuscular end plate depolarization leading to fasciculations and then flacid paralysis. CNS toxicity includes stimulation (including convulsions) followed by depression and paralysis of respiratory muscles.
- Varencline is a synthetic neuronal  $\alpha_4\beta_2$  nAChR partial agonist that decreases nicotine cravings and withdrawal symptoms. It is well absorbed and minimally metabolized with excellent bioavailability.

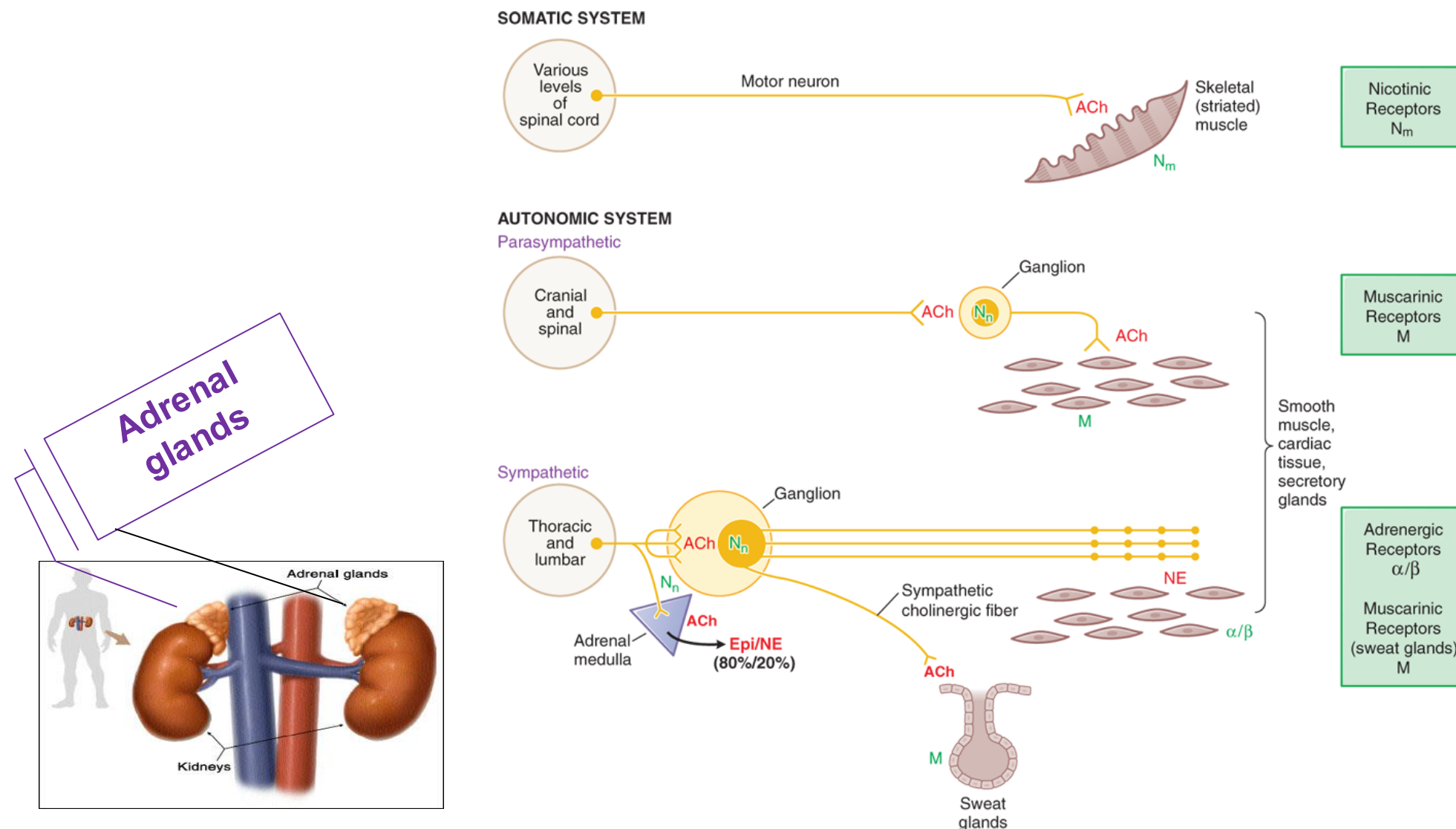
From: **3 The Autonomic Nervous System**, Figure 3.3  
Lippincott® Illustrated Reviews: Pharmacology, 8e, 2023

- The ANS regulates respiration, circulation, digestion, body temperature, metabolism, sweating, and the secretions of certain endocrine glands.
- **Parasympathetic craniosacral division of the ANS:**
  - ↓HR; ↓BP; constricts pupil (protects retina); stimulates bronchoconstriction and mucus secretion in the lungs; stimulates GI tract; empties urinary bladder/rectum
- **Sympathetic thoracolumbar division of the ANS:**
  - Constitutively active
  - “Fight or flight” response → ↑HR; ↑BP; blood flow shifts from skin/splanchnic to skeletal muscles; ↑blood glucose; bronchioles dilate; pupils dilate; CNS facilitation of response

- **Enteric NS:** self-contained (autonomous); GI functions

Cholinergic neurons stimulate smooth muscle contractions and secretions. Parasympathetic input and vaso-vagal reflexes are crucial for regulating numerous gut functions. Sympathetic regulation is inhibitory for maintaining homeostasis during stress or emergency situations.





Source: Laurence L. Brunton, Björn C. Knollmann:  
*Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e*:  
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Comparative features of somatic motor nerves and efferent nerves of the autonomic nervous system. The principal neurotransmitters, ACh and NE, are shown in red. The receptors for these transmitters, nicotinic (N) and muscarinic (M) cholinergic receptors,  $\alpha$  and  $\beta$  adrenergic receptors, are shown in green. Somatic nerves innervate skeletal muscle directly at a specialized synaptic junction, the motor end plate, where ACh activates  $N_m$  receptors. Autonomic nerves innervate smooth muscles, cardiac tissue, and glands. Both parasympathetic and sympathetic systems have ganglia, where ACh is released by the preganglionic fibers; ACh acts on  $N_n$  receptors on the postganglionic nerves. ACh is also the neurotransmitter at cells of the adrenal medulla, where it acts on  $N_n$  receptors to cause release of EPI and NE into the circulation. ACh is the dominant neurotransmitter released by postganglionic parasympathetic nerves and acts on muscarinic receptors. The ganglia in the parasympathetic system are near or within the organ being innervated, with generally a one-to-one relationship between pre- and postganglionic fibers. NE is the principal neurotransmitter of postganglionic sympathetic nerves, acting on  $\alpha$  or  $\beta$  adrenergic receptors. Autonomic nerves form a diffuse pattern with multiple synaptic sites. In the sympathetic system, the ganglia are generally far from the effector cells (e.g., within the sympathetic chain ganglia). Preganglionic sympathetic fibers may make contact with a large number of postganglionic fibers.

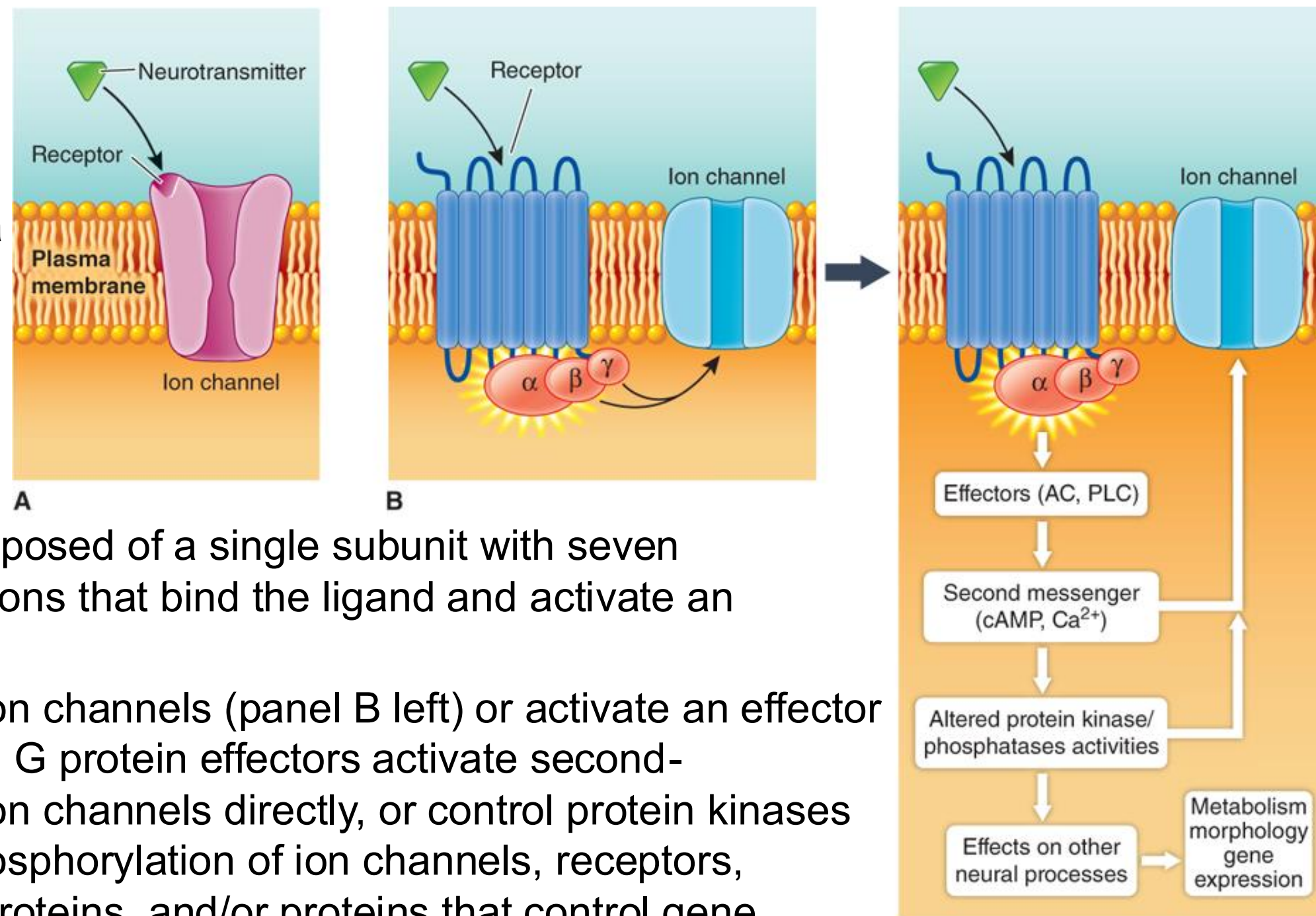
## Two types of neurotransmitter receptors.

A. Ionotropic receptors are ligand-gated channels that contain both a neurotransmitter (NT) binding domain and an integral ion channel domain. Most ionotropic receptors are composed of four or five subunits.

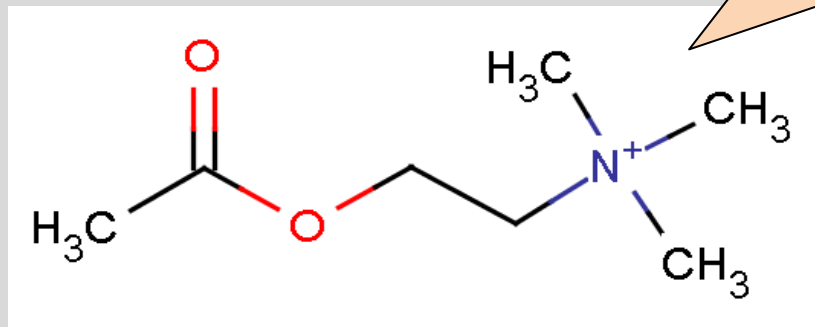
B. Metabotropic receptors are composed of a single subunit with seven membrane-spanning  $\alpha$ -helical regions that bind the ligand and activate an associated G protein.

G proteins can directly modulate ion channels (panel B left) or activate an effector (panel B right) such as AC or PLC. G protein effectors activate second-messengers, which can regulate ion channels directly, or control protein kinases and phosphatases that modulate phosphorylation of ion channels, receptors, metabolic enzymes, cytoskeletal proteins, and/or proteins that control gene expression.

AC, adenylyl cyclase; cAMP, cyclic adenosine monophosphate; PLC, phospholipase C.

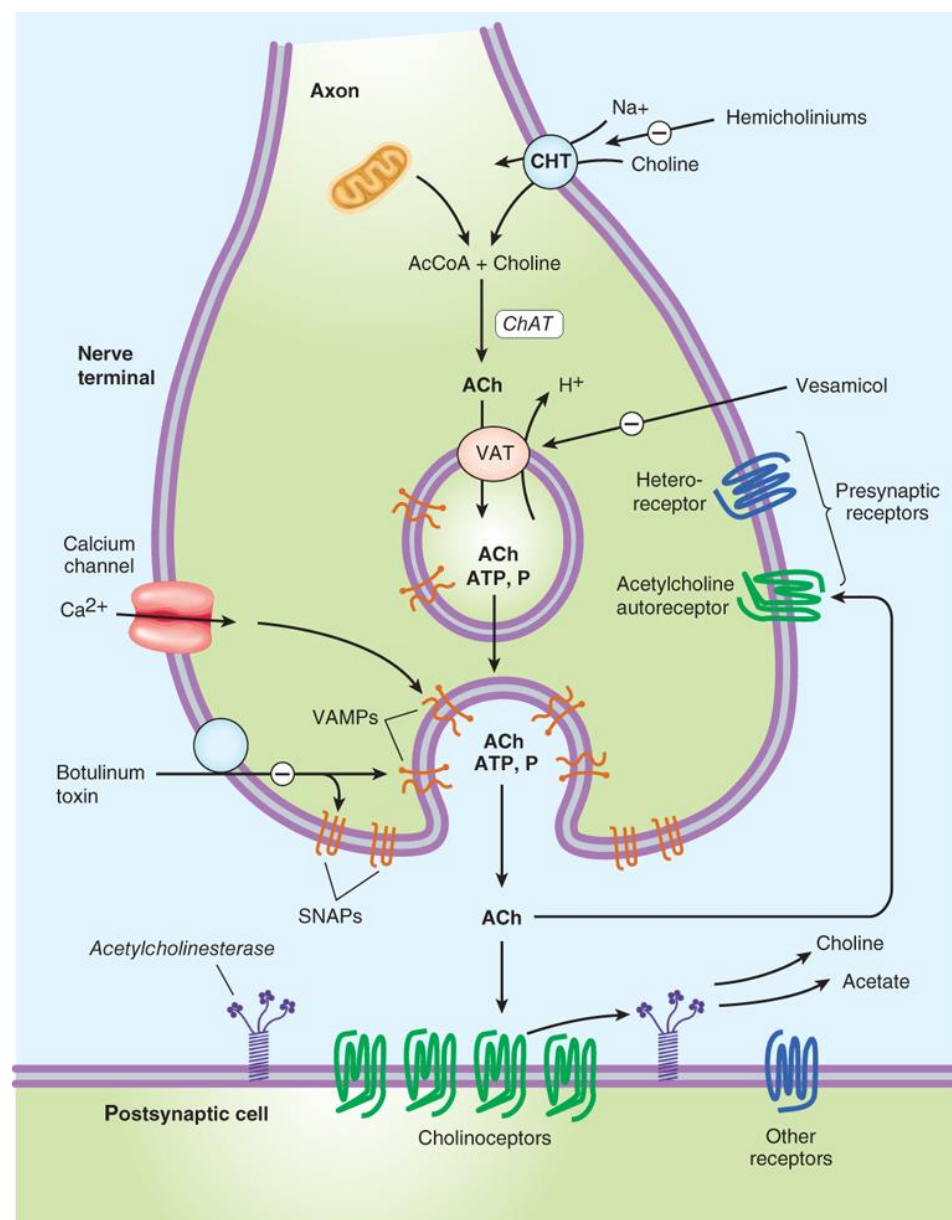


quaternary  
ammonium  
compound



# ACETYLCHOLINE

synthesis, storage, release, degradation



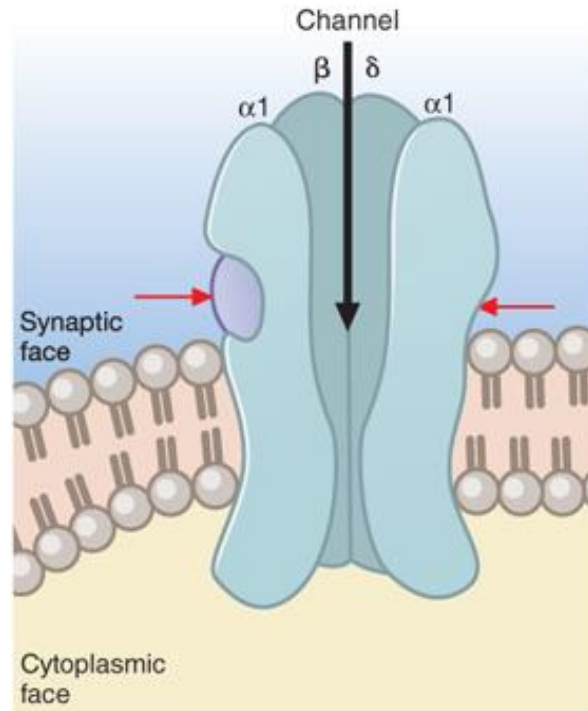
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1. Choline transport
2.  $\text{Ch} + \text{AcetylCoA} = \text{ACh}$
3. VAT → Transport of ACh into storage vesicle
4. Axon terminal depolarization →  $\text{Ca}^{2+}$  influx
5. Vesicle attachment and fusion
6. Exocytosis of ACh & cotransmitters into junctional cleft
7.  $\text{AChE} \rightarrow \text{Ch} + \text{acetate} \rightarrow \text{terminate action}$
8. Choline recycled; acetate diffuses away

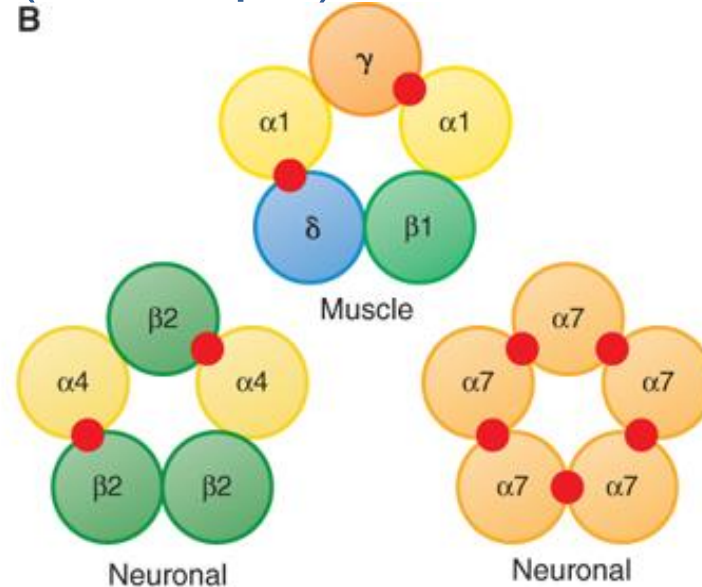
Schematic illustration of a generalized cholinergic junction (not to scale). Choline is transported into the presynaptic nerve terminal by a sodium-dependent choline transporter (CHT). This transporter can be inhibited by hemicholinium drugs. In the cytoplasm, acetylcholine is synthesized from choline and acetyl-CoA (AcCoA) by the enzyme choline acetyltransferase (ChAT). Acetylcholine (ACh) is then transported into the storage vesicle by a vesicle-associated transporter (VAT), which can be inhibited by vesamicol. Peptides (P), adenosine triphosphate (ATP), and proteoglycan are also stored in the vesicle. Release of transmitters occurs when voltage-sensitive calcium channels in the terminal membrane are opened, allowing an influx of calcium. The resulting increase in intracellular calcium causes fusion of vesicles with the surface membrane and exocytotic expulsion of acetylcholine and cotransmitters into the junctional cleft (see text). This step can be blocked by botulinum toxin. Acetylcholine's action is terminated by metabolism by the enzyme acetylcholinesterase. Receptors on the presynaptic nerve ending modulate transmitter release. SNAPs, synaptosomal nerve-associated proteins; VAMPs, vesicle-associated membrane proteins.

nicotinic ACh receptors are  
ion channels (ionotropic)

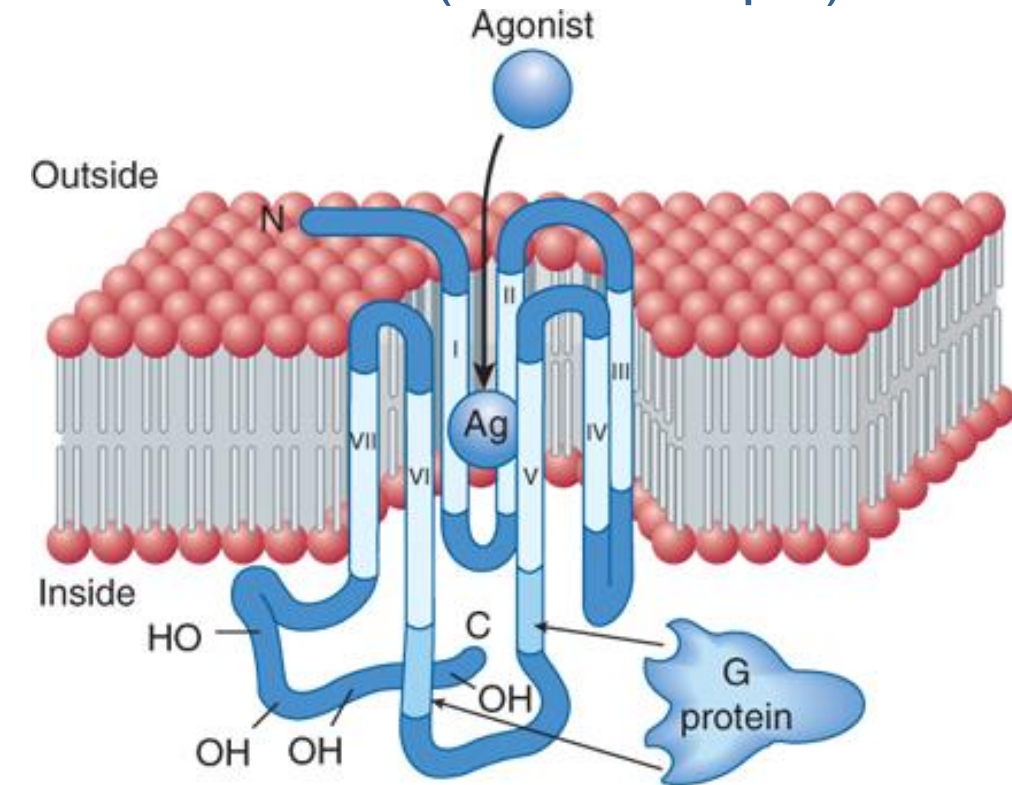
A



B



muscarinic ACh receptors are  
GPCRs (metabotropic)



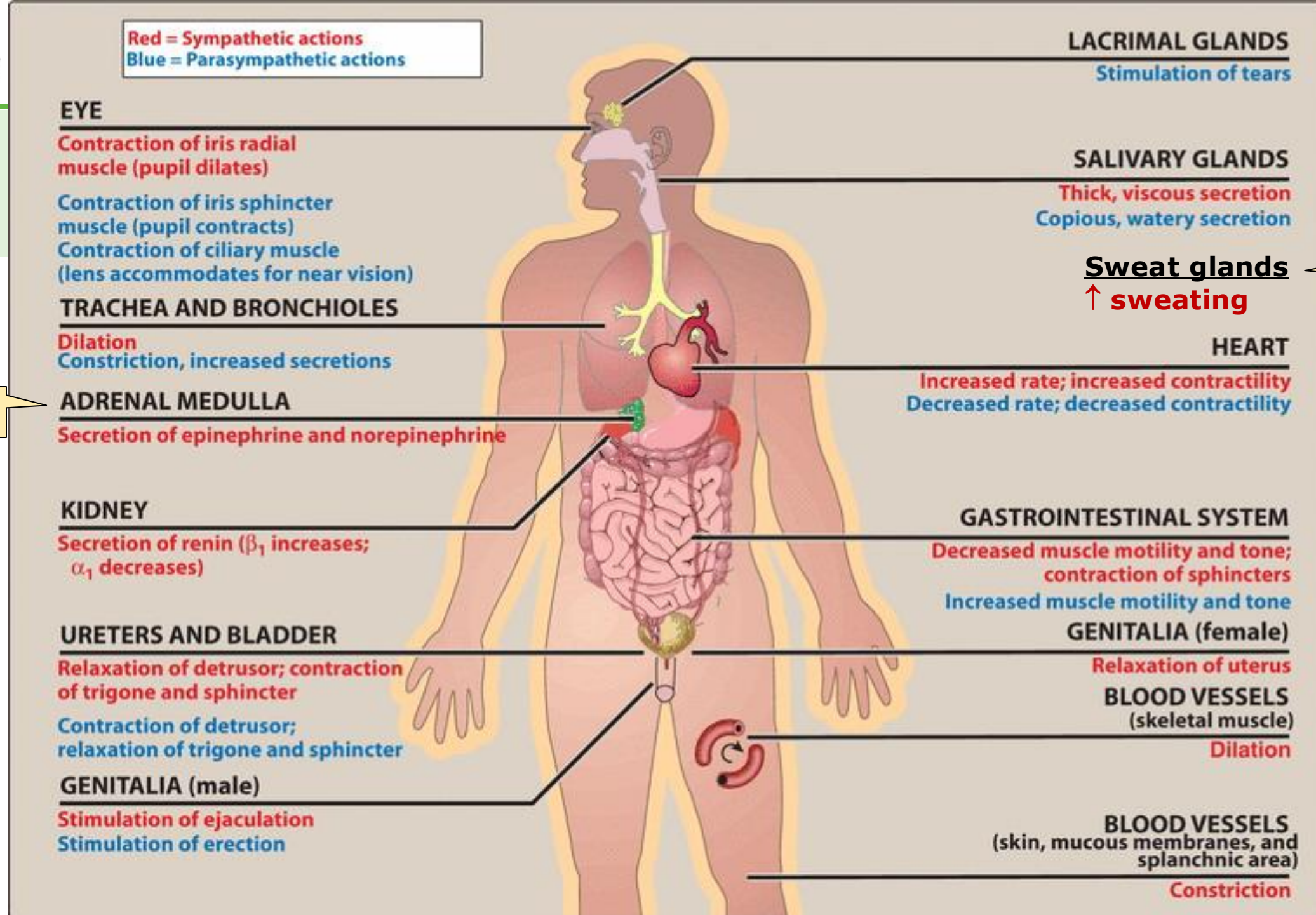
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Simultaneous binding of 2 ACh molecules at the interface between the  $\alpha$  subunits and the adjacent subunits significantly increases the channel opening probability, permitting one or more specific ions to diffuse through.

$N_N$  and  $N_M$  nicotinic Rs - ionotropic

$M_1, M_3, M_5$   $G_{q/11}$  PCRs  $\rightarrow$  PLC  $\rightarrow$   $IP_3$  & DAG  $\rightarrow$   
 $\uparrow Ca^{2+} + \uparrow PKC$ , respectively

$M_2, M_4$   $G_{i/o}$   $\rightarrow$  inhibits adenylyl cyclase  $\rightarrow$   $\downarrow cAMP$   
 $\rightarrow \uparrow K^+$  conductance and  $\downarrow Ca^{2+}$  conductance



$N_N$  receptor

$M_3$  receptor

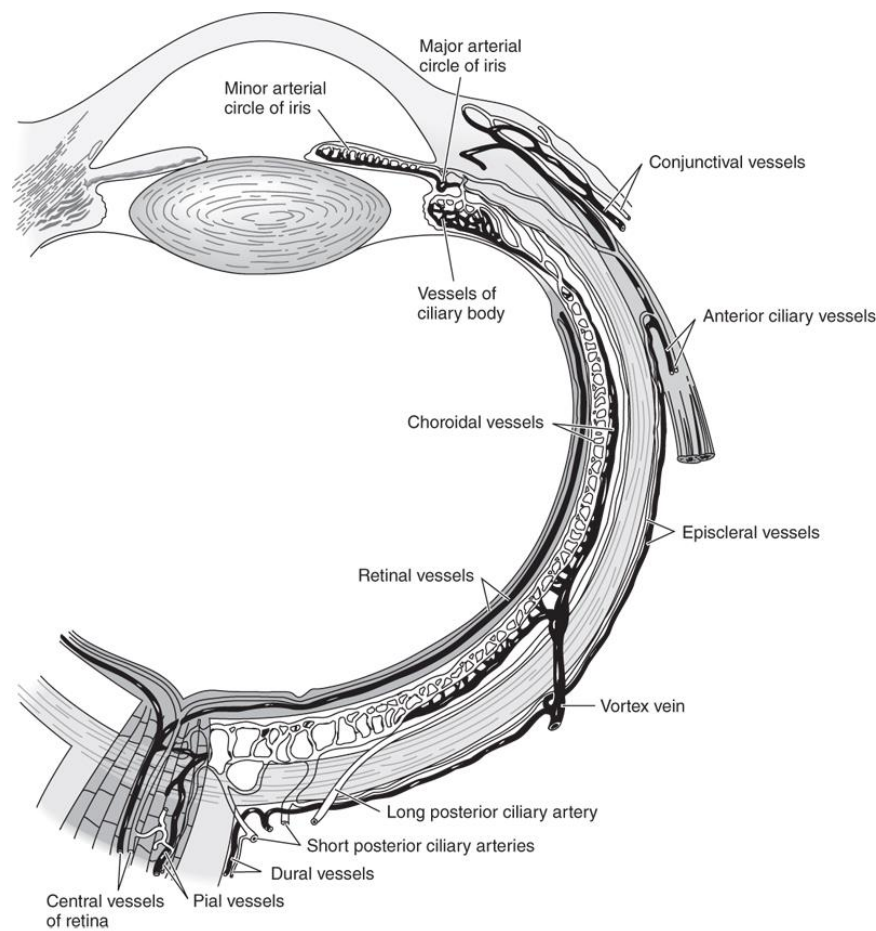
Nonneuronal receptors are also expressed on immune cells, vasculature ( $\beta_2$ ,  $M_3$ ), and bronchi ( $\beta_2$ )

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From: **3 The Autonomic Nervous System**, Figure 3.4

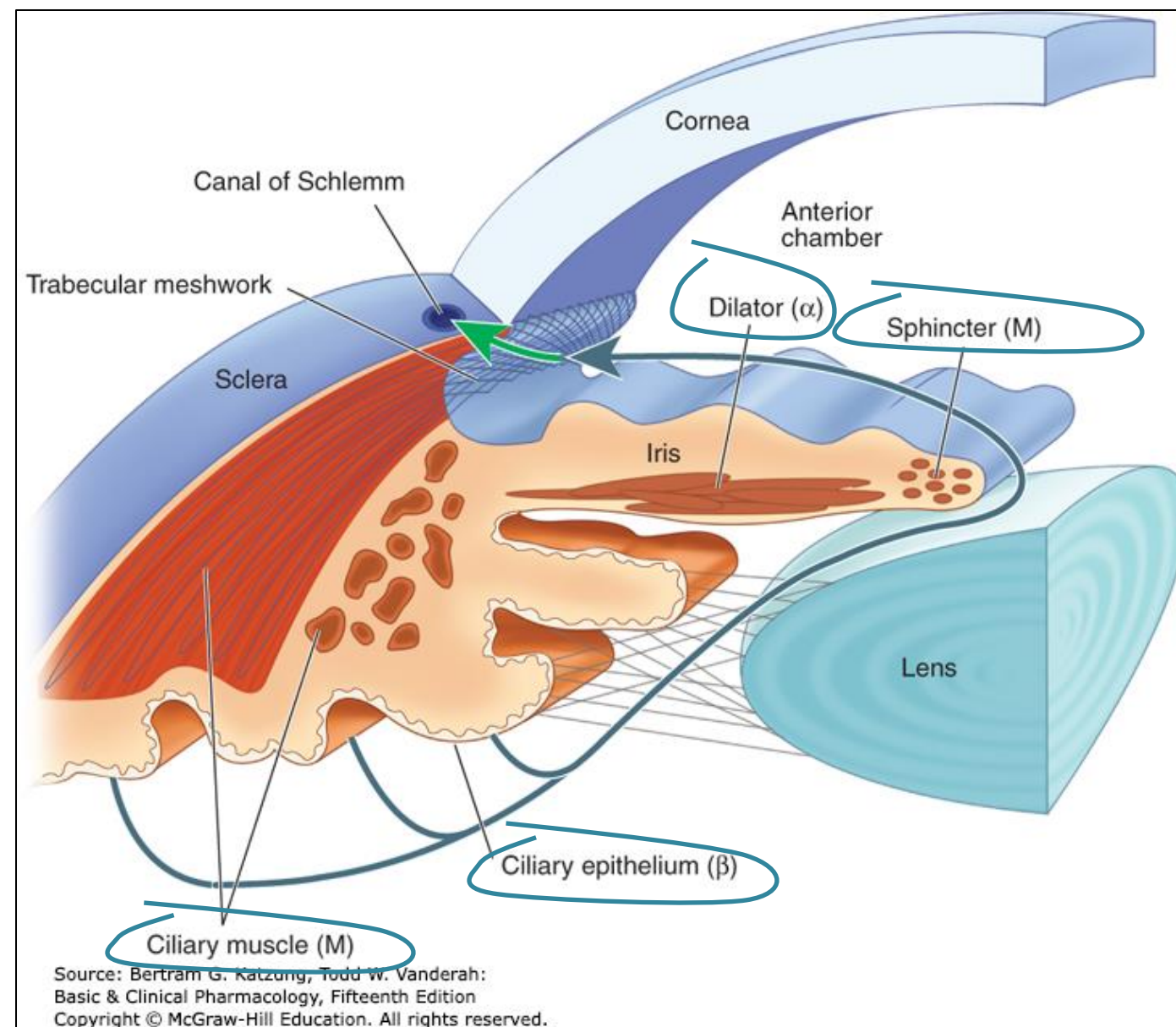
Actions of sympathetic and parasympathetic nervous systems on effector organs. Copyright © Wolters Kluwer

Slide adapted by LG



Source: P. Riordan-Eva, J. J. Augsburger: Vaughan & Asbury's General Ophthalmology, Nineteenth Edition  
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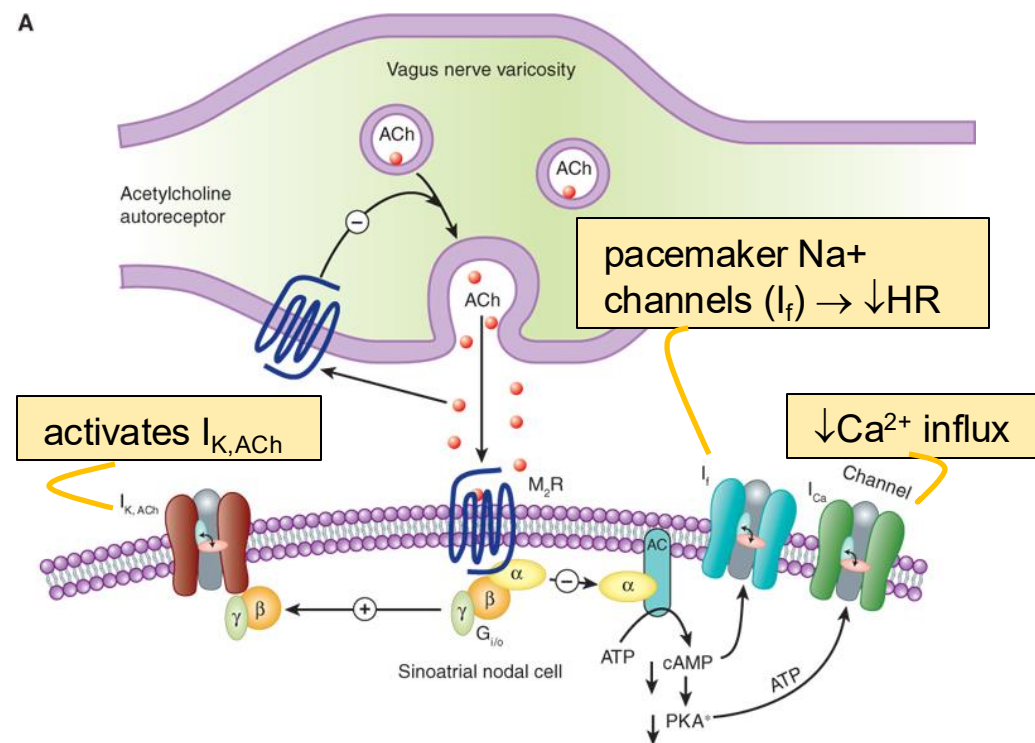
McGraw-Hill Chapter 1 Anatomy & Embryology of the Eye, Riordan-Eva P, Augsburger JJ. *Vaughan & Asbury's General Ophthalmology*, 19e; 2017. Figure 1-4



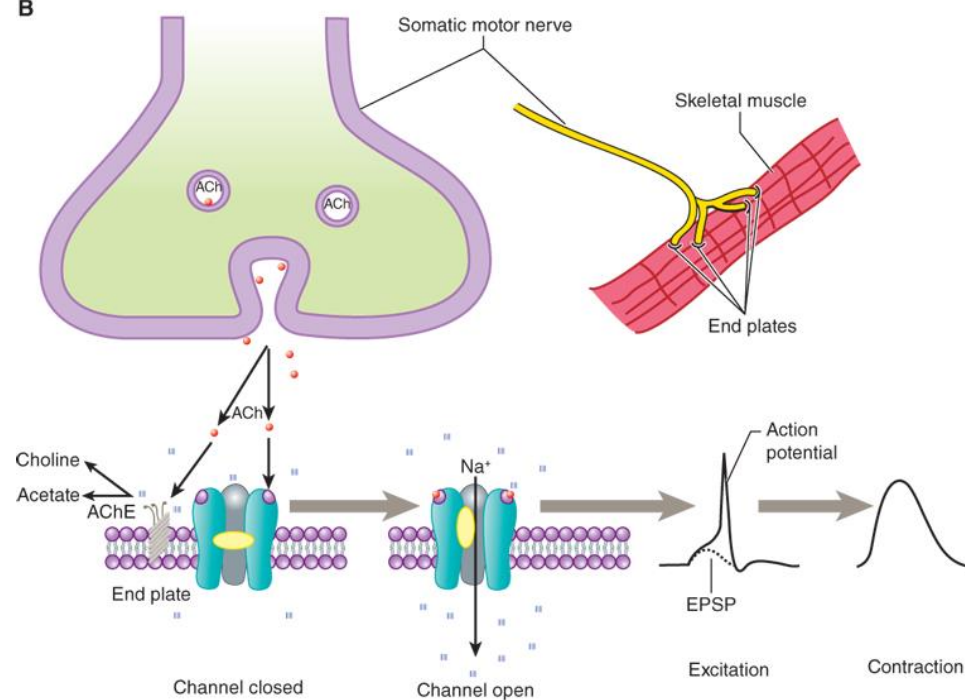
Source: Bertram G. Katzung, Todd W. Vanderah: Basic & Clinical Pharmacology, Fifteenth Edition  
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Structures of the anterior chamber of the eye. Tissues with significant autonomic functions and the associated ANS receptors ( $\alpha$ ,  $\beta$ , M) are shown in this schematic diagram. Aqueous humor is secreted by the epithelium of the ciliary body, flows into the space in front of the iris, flows through the trabecular meshwork, and exits via the canal of Schlemm (arrow). Blockade of the  $\beta$  adrenoceptors associated with the ciliary epithelium causes decreased secretion of aqueous. Blood vessels (not shown) in the sclera are also under autonomic control and influence aqueous drainage.

A



B



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Basic & Clinical Pharmacology, Sixteenth Edition  
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Citation: Chapter 7 Cholinoceptor-Activating & Cholinesterase-Inhibiting Drugs, Vanderah TW. *Katzung's Basic & Clinical Pharmacology*, 16<sup>th</sup> Edition; 2024, Figure 7-4. Available at: <https://accessmedicine.mhmedical.com/content.aspx?bookid=3382&sectionid=281747500>  
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## Muscarinic signaling in the heart / Nicotinic signaling in skeletal muscle

### Panel A:

Muscarinic transmission to the sinoatrial node in the heart **slows heart rate**.

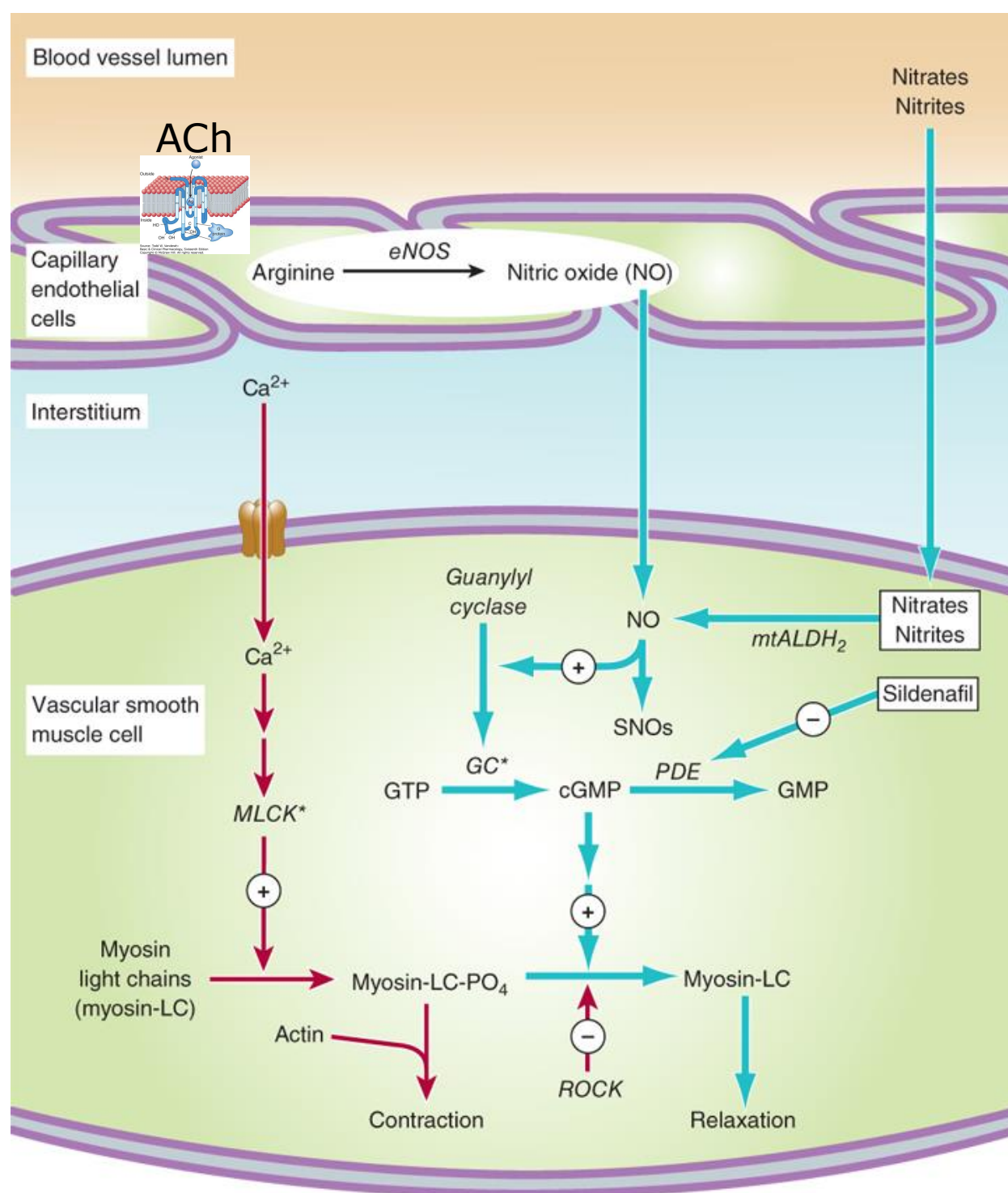
- Acetylcholine (ACh) released from a varicosity of a postganglionic cholinergic axon interacts with a sinoatrial node cell muscarinic receptor ( $M_2R$ ) linked via  $G_{i/o}$  to  $K^+$  channel opening, which causes hyperpolarization, and inhibition of cAMP synthesis.
- Reduced cAMP shifts the voltage-dependent opening of **pacemaker channels ( $I_f$ )** to more negative potentials and reduces the phosphorylation and availability of **L-type  $Ca^{2+}$  channels ( $I_{Ca}$ )**.
- Released ACh also acts on an axonal muscarinic autoreceptor to cause inhibition of ACh release (autoinhibition).
- Muscarinic transmission also slows conduction through the AV node and reduces the force of contraction. (These effects are not shown in the figure.)

### Panel B:

Nicotinic transmission at the skeletal neuromuscular junction creates an end plate potential (if strong enough).

- ACh released from the motor nerve terminal interacts with subunits of the pentameric nicotinic receptor to open it, allowing  $Na^+$  influx to produce an excitatory postsynaptic potential (EPSP).
- The EPSP depolarizes the muscle membrane, generating an action potential, and triggering contraction.
- Acetylcholinesterase (AChE) in the extracellular matrix hydrolyzes (inactivates) ACh

Slide adapted by LG



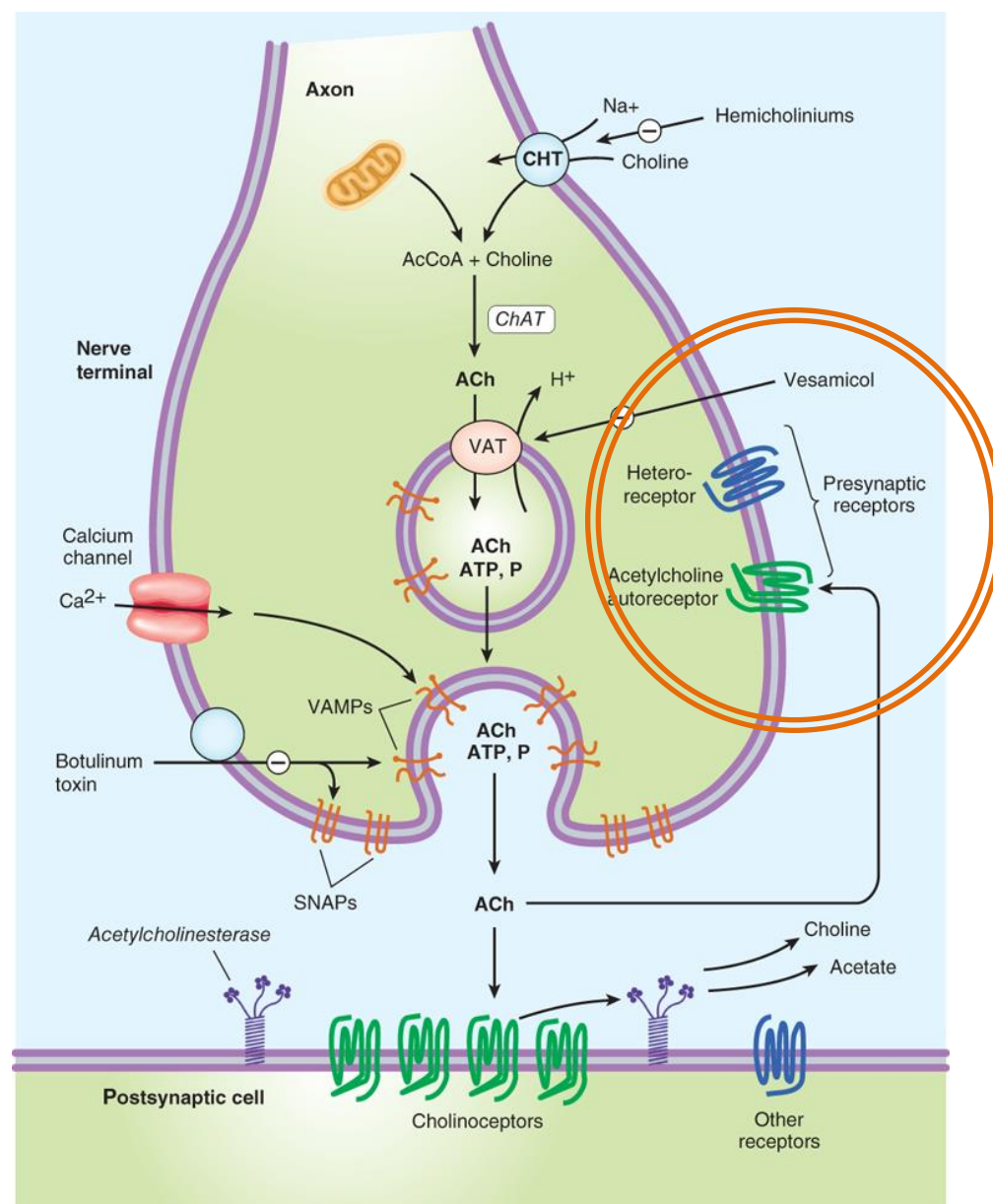
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## Mechanism of nitric oxide (NO)-mediated vasodilation

- Endothelial nitric oxide synthase (eNOS) is activated by fluid sheer stress, numerous agonists, interactions with various substrates, cofactors, regulatory proteins, and more.
- NO is released, diffuses into vascular smooth muscle cells, and induces relaxation.

Effect of acetylcholine intravenous infusion (investigative use – not used in clinical practice)

- ACh IV infusion activates nonneuronal M<sub>3</sub> receptors on vascular endothelium, which stimulates eNOS and NO synthesis.
- Small IV doses cause vasodilation with a reduction in blood pressure, often accompanied by a reflex increase in heart rate
- (ACh action is not specifically shown on the figure.)



Source: Bertram G. Katzung, Todd W. Vanderah:  
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- **Local integration:** Feedback by modulatory autoreceptors and heteroreceptors on presynaptic nerve membranes regulate neurotransmitter release.

- **Systemic reflexes:**

Homeostatic mechanisms regulate blood pressure, gastrointestinal motility, bladder tone, airway smooth muscle, and other processes.

For example, a drop in blood pressure is detected by the baroreceptors in the carotid arteries and aortic arch → signals the sympathetic nervous system in the CNS → increases heart rate and force of contraction → heart pumps more blood + vasoconstriction → increase blood pressure

- ❖ Remember, the nervous system is an interconnected, intricate network that ensures the body maintains homeostasis and appropriate responses to various stimuli.

Schematic illustration of a generalized cholinergic junction (not to scale). Choline is transported into the presynaptic nerve terminal by a sodium-dependent choline transporter (CHT). This transporter can be inhibited by hemicholinium drugs. In the cytoplasm, acetylcholine is synthesized from choline and acetyl-CoA (AcCoA) by the enzyme choline acetyltransferase (ChAT). Acetylcholine (ACh) is then transported into the storage vesicle by a vesicle-associated transporter (VAT), which can be inhibited by vesamicol. Peptides (P), adenosine triphosphate (ATP), and proteoglycan are also stored in the vesicle. Release of transmitters occurs when voltage-sensitive calcium channels in the terminal membrane are opened, allowing an influx of calcium. The resulting increase in intracellular calcium causes fusion of vesicles with the surface membrane and exocytotic expulsion of acetylcholine and cotransmitters into the junctional cleft (see text). This step can be blocked by botulinum toxin. Acetylcholine's action is terminated by metabolism by the enzyme acetylcholinesterase. Receptors on the presynaptic nerve ending modulate transmitter release. SNAPs, synaptosomal nerve-associated proteins; VAMPs, vesicle-associated membrane proteins.

## Check your knowledge

- What is the definition of cholinergic receptors?
- 
- What are the cholinergic receptor subtypes?
- 
- Where in the body are cholinergic receptors located?
-

## Therapeutic uses of muscarinic agonists by acting on M3 receptors

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetylcholine | Instillation, anterior chamber: Induction of miosis for ophthalmologic surgery                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                        |
| Methacholine  | <p>Inhalation: Bronchoprovocation diagnostic challenge test for bronchial airway hyper-reactivity in patients suspected of having asthma</p> <p>➤ Rationale: Airway hyperresponsiveness may be the sole evidence of airway dysfunction/asthma.</p> <p>Duration 30-45 minutes unless reversed by rapid-acting bronchodilator</p> <p>Reversal agent: <math>\beta</math>2-adrenergic agonist, i.e., albuterol (reversal occurs in ~5 min)</p> |                                                                                                                                                                        |
| Carbachol     | Topical eye drops: Glaucoma; induction of miosis during ophthalmologic surgery                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                        |
| Bethanechol   | Oral: Urinary retention due to neurogenic bladder<br>acute post-operative or post-partum urinary retention                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                        |
| Pilocarpine   | <p>Topical eye drops:</p> <p>Induction of miosis</p> <p>Acute angle-closure glaucoma</p> <p>Chronic open-angle glaucoma</p>                                                                                                                                                                                                                                                                                                                | <p>Oral: Sialagogue</p> <p>Symptomatic treatment of dry mouth (xerostomia) caused by Sjögren's syndrome or salivary gland hypofunction resulting from radiotherapy</p> |
| Cevimeline    | Oral: Sialagogue for the symptomatic treatment of dry mouth (xerostomia) caused by Sjögren's syndrome or salivary gland hypofunction resulting from radiotherapy.                                                                                                                                                                                                                                                                          |                                                                                                                                                                        |

## Adverse effects of muscarinic agonists by acting on M3 receptors

(not a complete list)

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                           |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| All mAChR agonists | <p>Systemic effects: bradycardia, hypotension with reflex tachycardia, dyspnea</p> <p>➤ Tachycardia increases cardiac oxygen demand and increases blood pressure.</p> <p>Caution: recent MI, uncontrolled hypertension, aortic aneurysm, urinary obstruction, GI obstruction, any condition that could be adversely affected by a cholinergic agent (respiratory disease; peptic ulcer disease; hyperthyroidism; GI, biliary, urinary obstruction)</p> |                                                                                                                           |
| Acetylcholine      | Ophthalmic: cloudy vision, corneal edema                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                           |
| Methacholine       | Inhalation: bronchospasm (can be severe), Caution: Baseline FEV1 <60%                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                           |
| Carbachol          | Ophthalmic: blurred vision, corneal edema, headache                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                           |
| Bethanechol        | Oral: Blood pressure decrease with tachycardia, bronchoconstriction, diaphoresis, lacrimation (tearing), salivation, nausea, abdominal cramps, urinary urgency,                                                                                                                                                                                                                                                                                        |                                                                                                                           |
| Pilocarpine        | Topical eye drops: blurred vision, impaired vision, accommodation disturbance                                                                                                                                                                                                                                                                                                                                                                          | <p>Oral: dizziness, headache, asthenia (feeling of weakness)</p> <p>Peripheral cholinergic effects (like bethanechol)</p> |
| Cevimeline         | Oral: CNS and peripheral cholinergic effects typical of the class                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                           |

## Check your knowledge

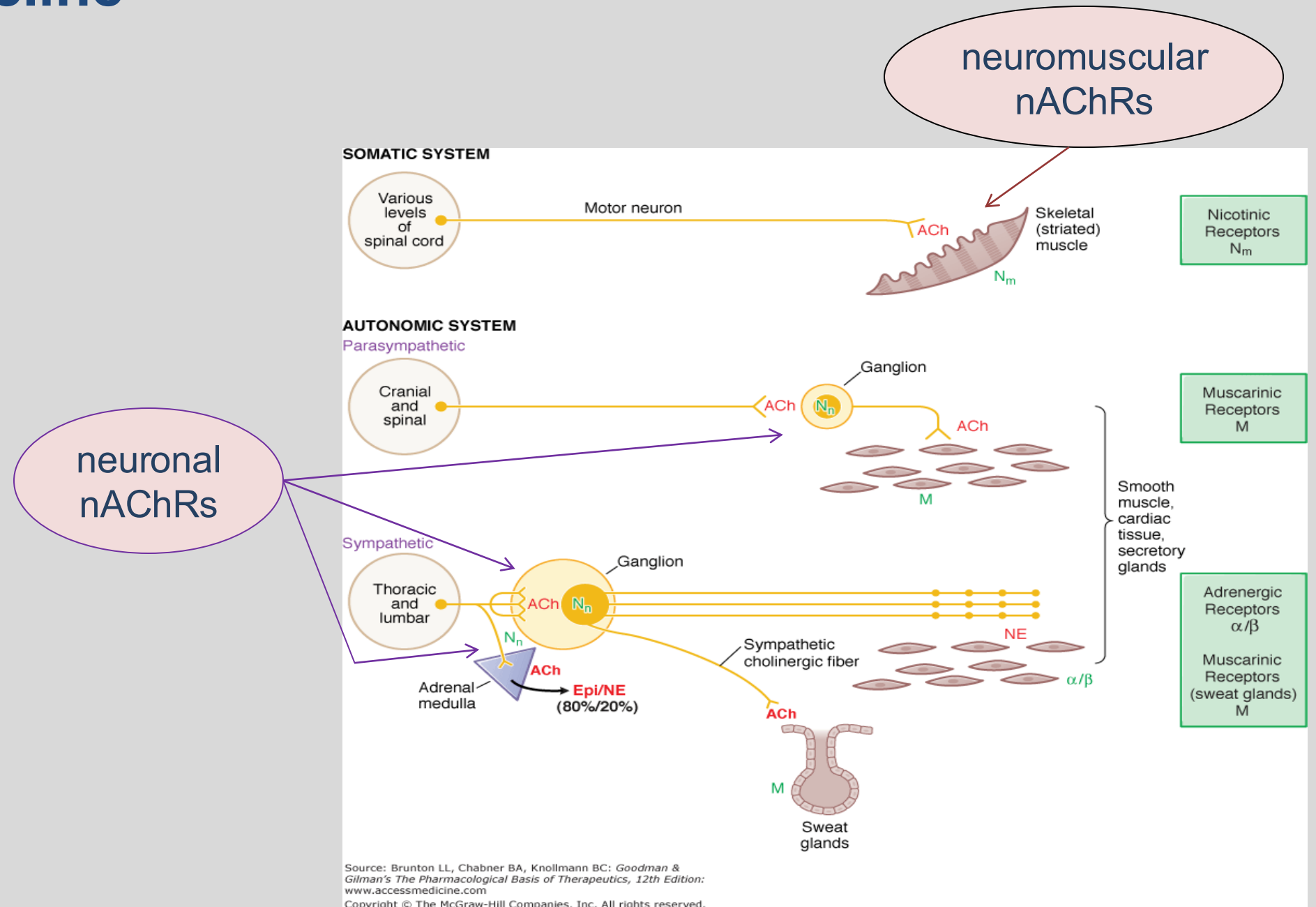
- Acetylcholinesterase can hydrolyze 5000 acetylcholine molecules per second. Why, then, is the duration of action of ACh instilled in the anterior chamber as long as 6 hours?
- 
- An asymptomatic 6-year-old patient suspected of having asthma has a positive bronchoprovocation test using an orally inhaled cholinergic agent. What are the mechanism of action and effects of the administered drug?
- 
- What are the mechanism-related side effects of muscarinic receptor agonists?
-

# Direct-Acting Nicotinic Receptor Agonists

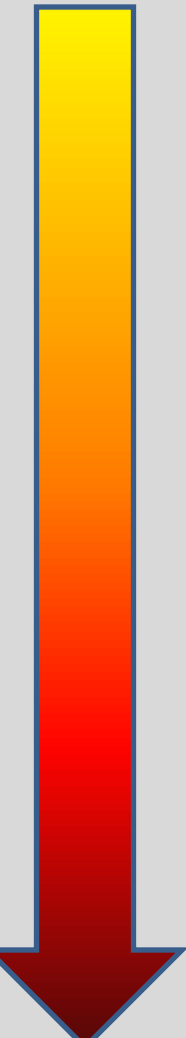
## Nicotine and Varenicline



Nicotine is an alkaloid of *Nicotiana glauca*, a plant of the nightshade family indigenous to the Americas and other areas.  
Painting by William Rossin posters



## Acute Nicotine Toxicity (concentration-dependent)

- 
- Cholinergic excess
  - Central stimulation
  - Hypertension, tachycardia  
(sympathetic stimulation at  
ganglia and adrenal medulla)  
followed by  
hypotension, bradycardia  
(depolarizing block of  
autonomic ganglia)
  - **Convulsions**
  - **Death by respiratory arrest**  
central respiratory center depression  
and respiratory muscle paralysis

### Depolarizing block mechanism:

- Initial stimulation of central, autonomic, and neuromuscular nicotinic receptors
- Prolonged agonist occupation of receptors causes persistent depression



- **depolarizing blockade**

### **Blockade at motor end plate →**

- **muscle fasciculations**
- **muscle weakness / flaccid paralysis**
- **respiratory muscle paralysis**

## Check your knowledge

- Where are nicotinic receptors located?
- 
- What type of receptor are nicotinic receptors?
- 
- A healthy adult patient wishes to quit smoking? What two drugs in this category can be recommended?
- 
- What are the differences between these two drugs?
- 
- What are the symptoms/signs of acute nicotine toxicity?
-

## Summary: Direct-Acting Cholinomimetics

|                                                |                                                                            |                                                                                                                                                       |
|------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Acetylcholine</b><br>intraocular            | Miotic agent,<br>adjunct ophthalmic surgery                                | <ul style="list-style-type: none"> <li>• Rapid acting</li> </ul>                                                                                      |
| <b>Methacholine</b><br>inhalation              | Dx test:<br>bronchial hyperreactivity                                      | <ul style="list-style-type: none"> <li>• Inhalation only</li> <li>• Severe bronchoconstriction</li> <li>• Medical supervision</li> </ul>              |
| <b>Carbachol</b><br>ophthalmic drops           | Glaucoma<br>Adjunct ophthalmic surgery                                     | <ul style="list-style-type: none"> <li>• Carbachol ophthalmic use only</li> <li>• Nicotinic + muscarinic properties when used systemically</li> </ul> |
| <b>Bethanechol</b><br>Oral; variable abs       | Stimulate urinary motility                                                 | <ul style="list-style-type: none"> <li>• Generalized muscarinic effects</li> <li>• Lacks nicotinic actions</li> </ul>                                 |
| <b>Pilocarpine</b><br>ophthalmic drops<br>oral | Angle-closure glaucoma<br>Open-angle glaucoma<br>Xerostomia (oral)         | <ul style="list-style-type: none"> <li>• Profuse sweating, salivation</li> <li>• CNS effects</li> </ul>                                               |
| <b>Cevimeline</b><br>oral                      | Xerostomia of Sjogren's syndrome<br>or radiation damage of salivary glands | <ul style="list-style-type: none"> <li>• M<sub>3</sub> affinity</li> <li>• Hepatic metabolism (CYP2D6,3A4)</li> </ul>                                 |

## Check your knowledge

- What is the definition of cholinergic receptors?
- Acetylcholine is the neurotransmitter for nicotinic and muscarinic receptors.
- What are the cholinergic receptor subtypes?
- Nicotinic receptors are ligand-gated ionotropic receptors. Sodium, potassium, and sometimes calcium pass through them. Muscarinic receptors  $M_1$ ,  $M_2$ ,  $M_3$ ,  $M_4$ ,  $M_5$  are G protein-coupled receptors located on effector cells.  $M_{1,3,5}$  are postsynaptic GqPCRs.  $M_{2,4}$  are GiPCRs, auto- and heteroreceptors.
- Where in the body are cholinergic receptors located?
- Nicotinic receptors in the periphery are located at all ANS ganglia and the adrenal medulla. The muscarinic receptors are distributed in the heart, blood vessels, bronchi, GI tract, urinary tract, and salivary, lacrimal, mucous, and sweat glands. Nicotinic and all types of muscarinic receptors are distributed across various regions of the CNS and non-neuronal cells, such as immune cells (N and M) and endothelial cells ( $M_3$ ).

## Check your knowledge

- Acetylcholinesterase can hydrolyze 5000 acetylcholine molecules per second. Why, then, is the duration of action of ACh instilled in the anterior chamber as long as 6 hours?
- In neuron synapses, AChE is highly concentrated, which is necessary for precise regulation of ACh activity, but there is a lower concentration of AChE in the anterior chamber of the eye→ reduced activity→ longer duration of ACh action.
- An asymptomatic 6-year-old patient suspected of having asthma has a positive bronchoprovocation test using an orally inhaled cholinergic agent. What are the mechanism of action and effects of the administered drug?
- Administration of methacholine by oral inhalation activates M3 receptors in bronchial smooth muscle, causing bronchoconstriction. In patients with pre-existing bronchial hyperreactivity, bronchoconstriction will be provoked at lower methacholine doses. A patient with a negative bronchoprovocation test is unlikely to have asthma. Bronchoprovocation testing may be useful for assessment of disease severity, prognosis, and plan of asthma treatment.
- What are the mechanism-related side effects of muscarinic receptor agonists?
- GI: sialorrhea, dyspepsia, abdominal upset, diarrhea, UT: urinary frequency, urgency, incontinence, CV: palpitations, Eye: blurred vision, visual disturbance, CNS: drowsiness, asthenia

## Check your knowledge

- Where are nicotinic receptors located?
- CNS, ANS ganglia (both PNS and SNS), chromaffin cells of adrenal medulla
- What type of receptor are nicotinic receptors?
- Ligand-gated ionotropic ion channels activated by ACh binding to sites on  $\alpha$  subunits adjacent to neighboring subunits. Opening of the channels increases permeability to  $\text{Na}^+$ ,  $\text{K}^+$ , and in some channels  $\text{Ca}^{2+}$ .
- A healthy adult patient wishes to quit smoking? What two drugs in this category can be recommended?
- Nicotine, available in several different products, and oral varenicline.
- What are the differences between these two drugs?
- Nicotine is a full agonist. Varenicline is a partial agonist at  $\alpha 4\beta 2$  neuronal nAChR. Both drugs help reduce cravings and withdrawal symptoms.
- What are the symptoms/signs of acute nicotine toxicity?
- Cholinergic excess (stimulation of receptors on effector cells), CNS stimulation sympathetic excess (effect at ganglia) followed by hypotension and bradycardia (depolarizing block of autonomic ganglia), convulsions, death by respiratory arrest